

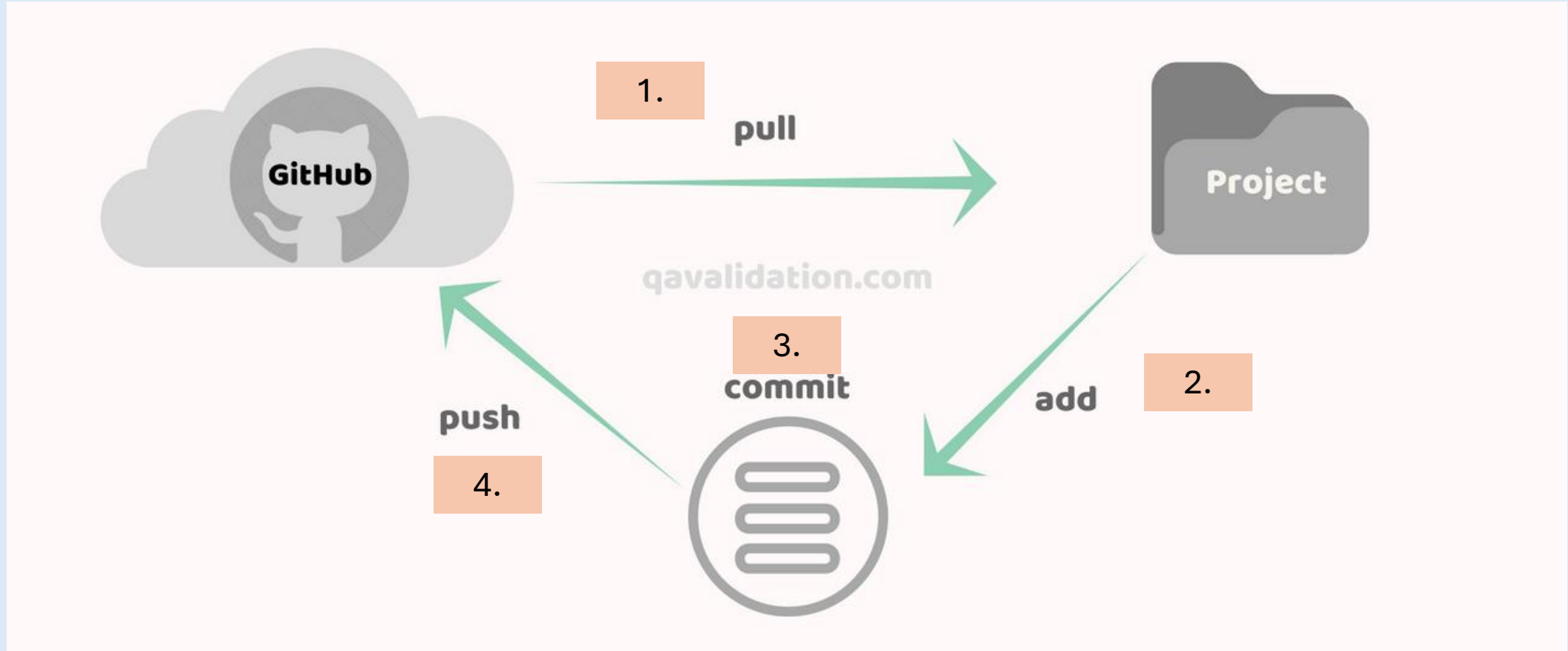
Coding Together

Human-Centered Design

Collaboration

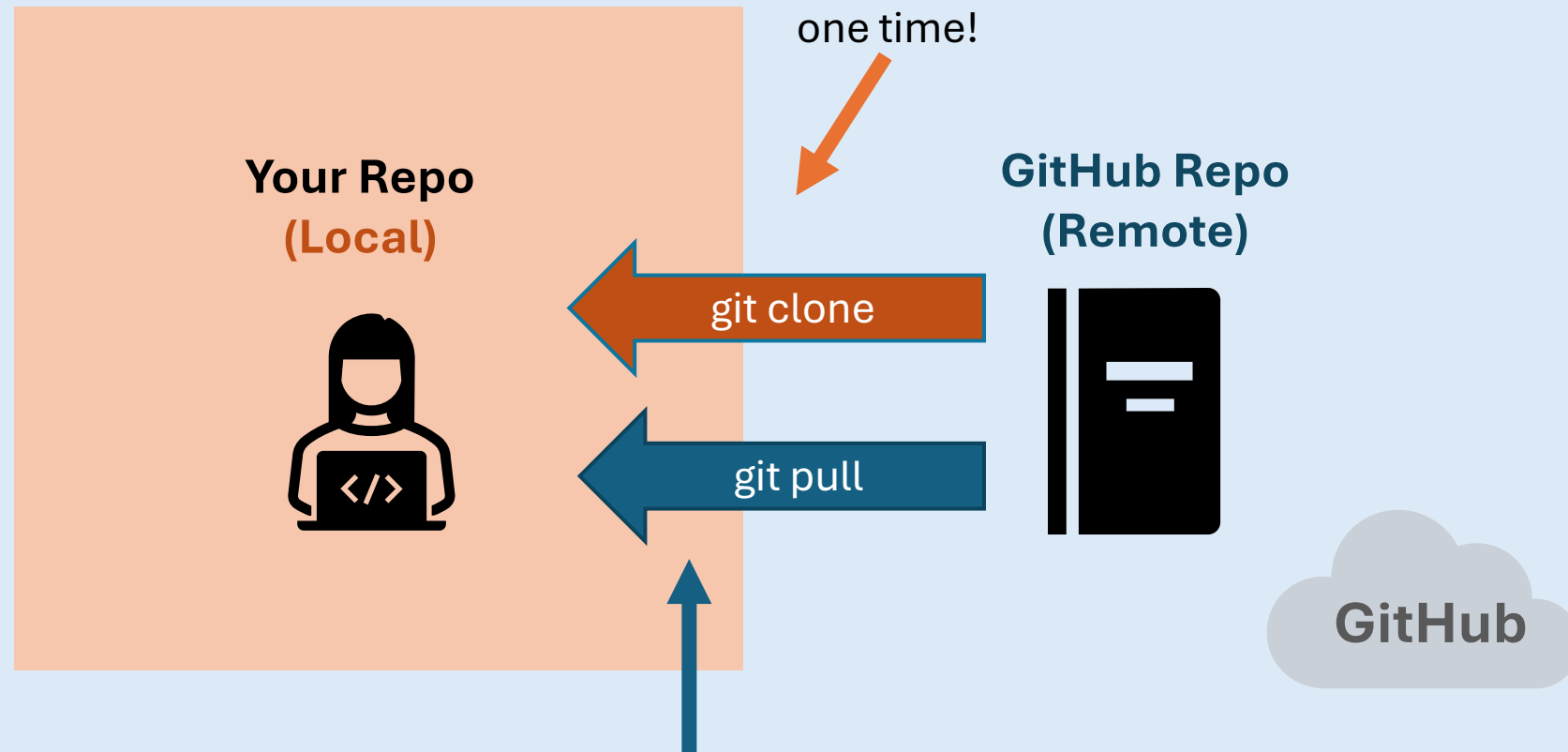
Albert Kochaphum
Lab 5 - Extra | July 24th, 2026

The git process



1. Pull changes
2. Add changes
3. Commit changes (with a message)
4. Push changes

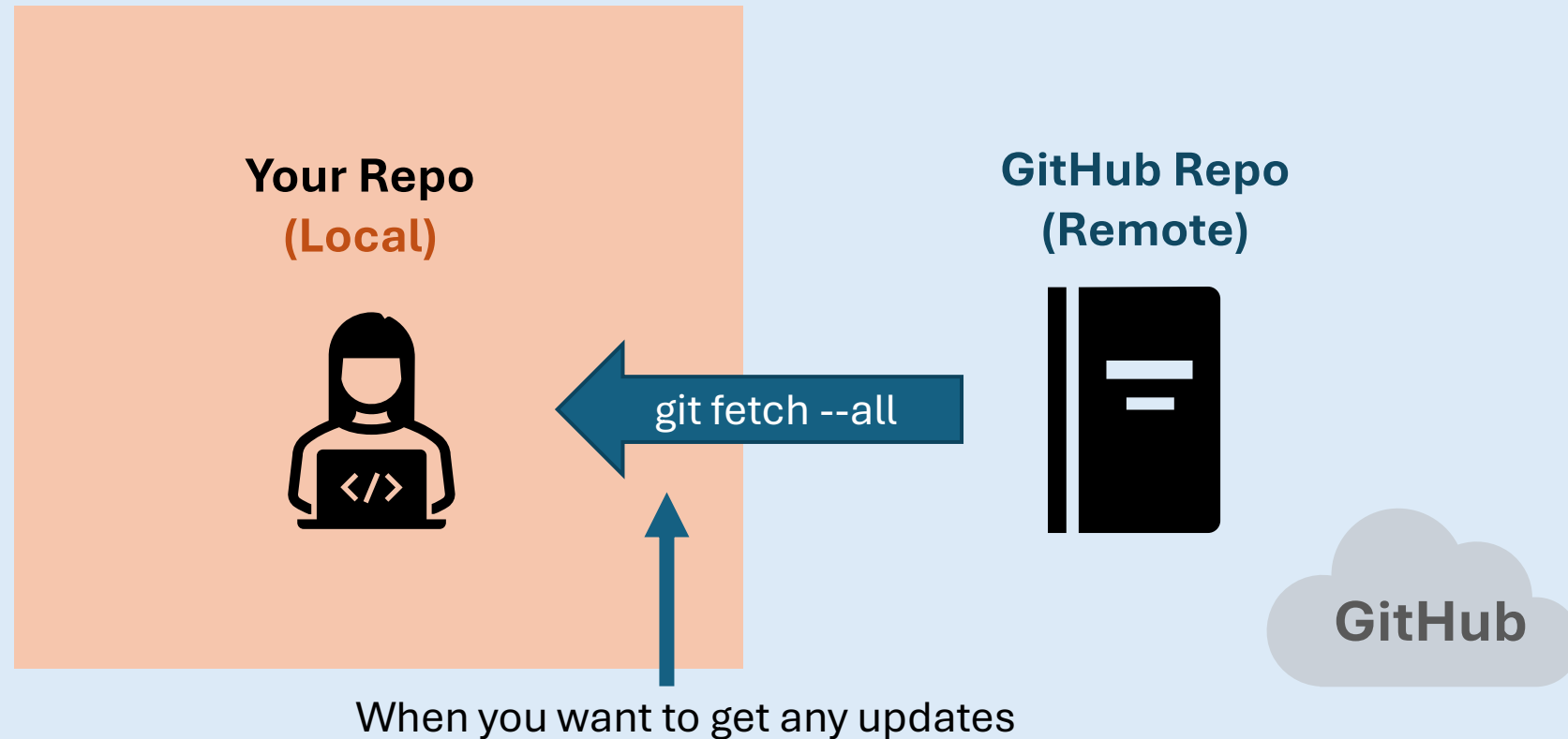
git cloning/git pull



When you want to get updates

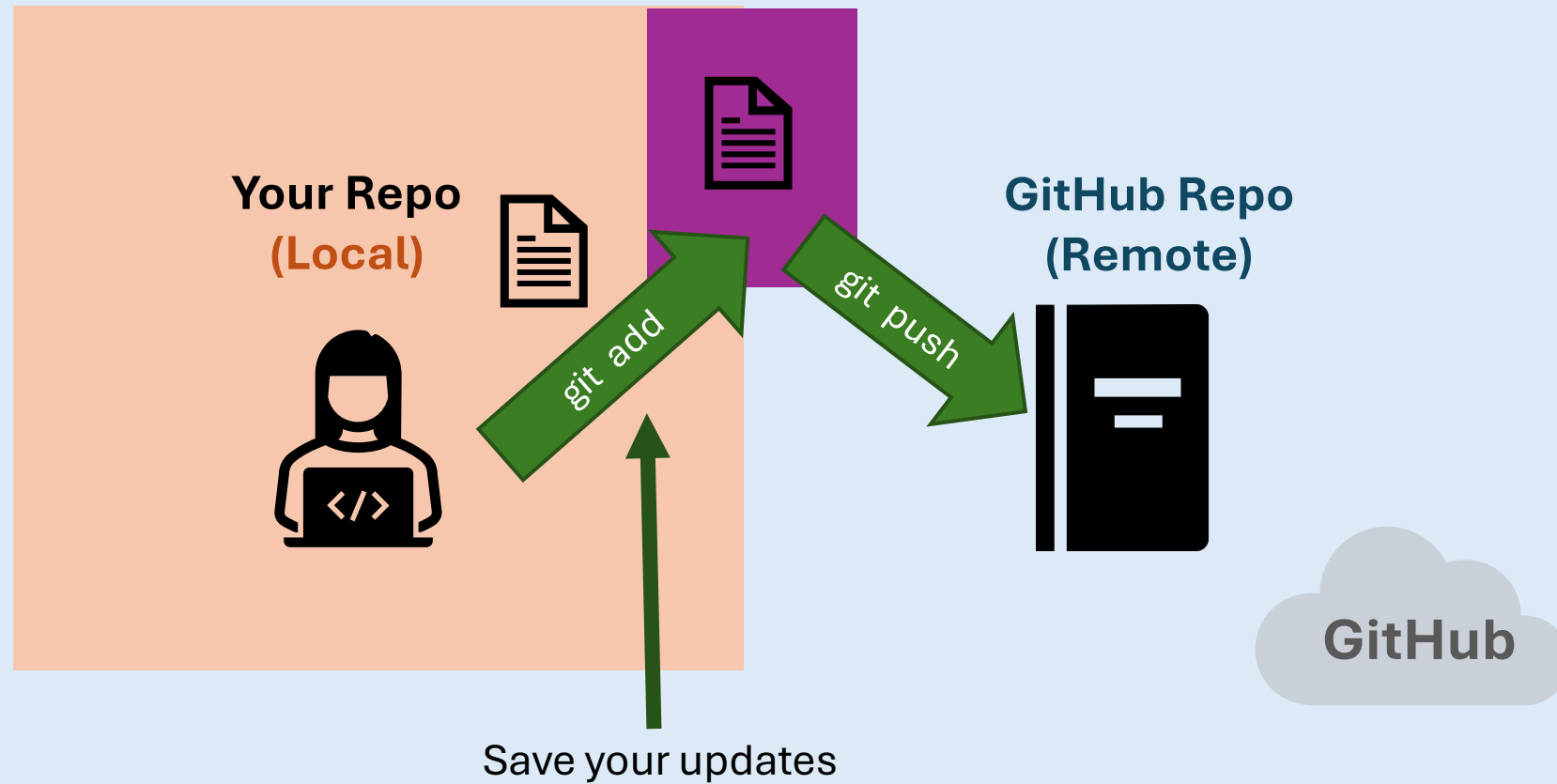
“Getting code”

git fetch --all



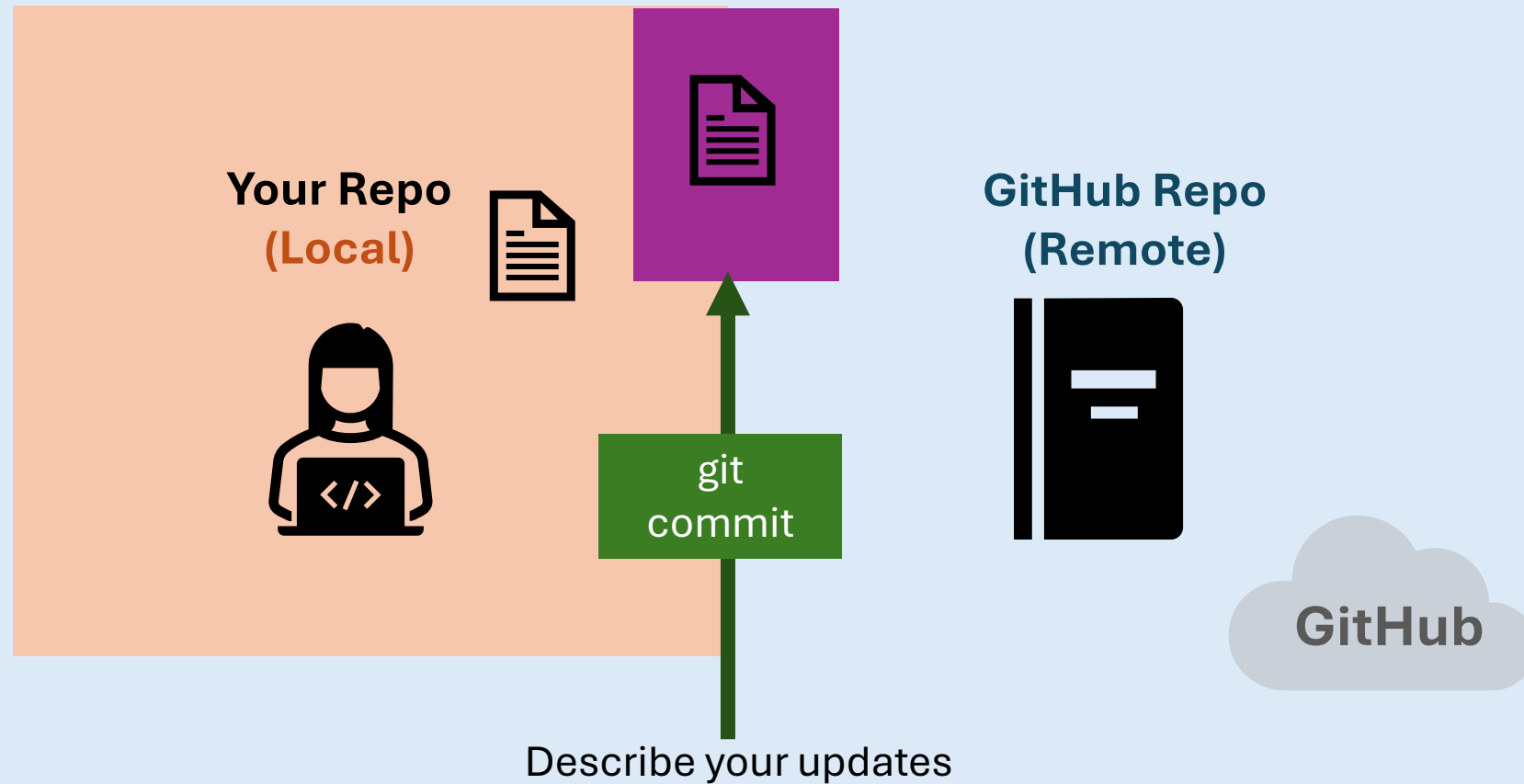
“get new code” from **branches**

git add



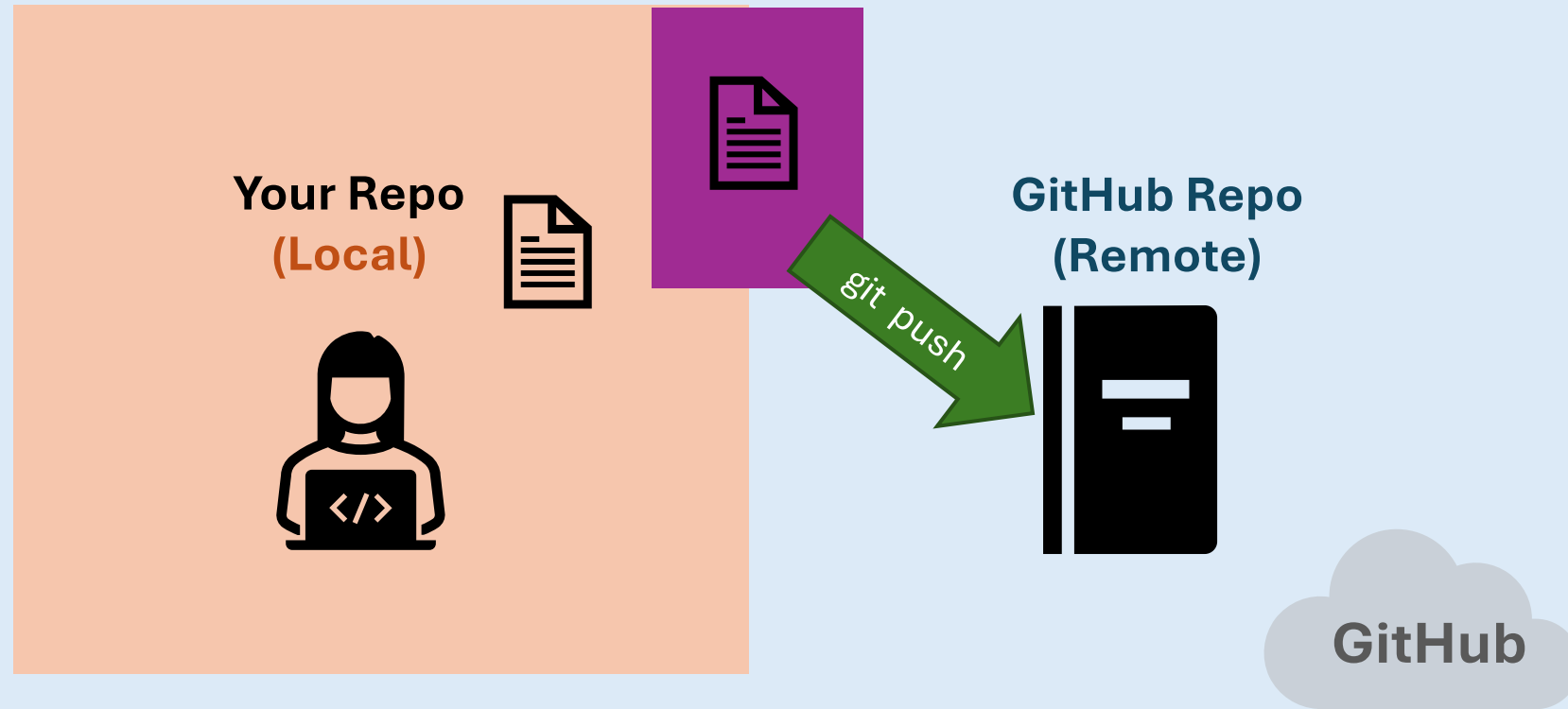
Choose what to save

git commit



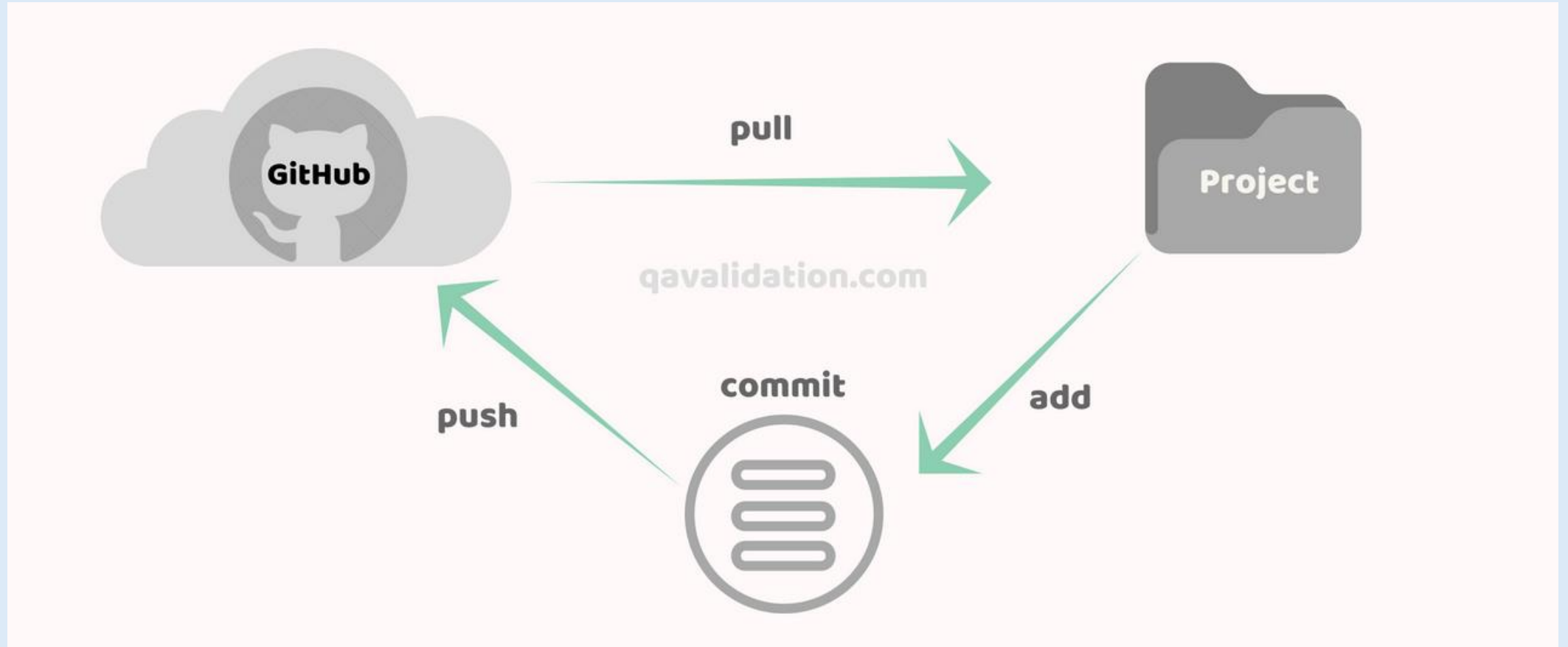
Add a message to your save

git push



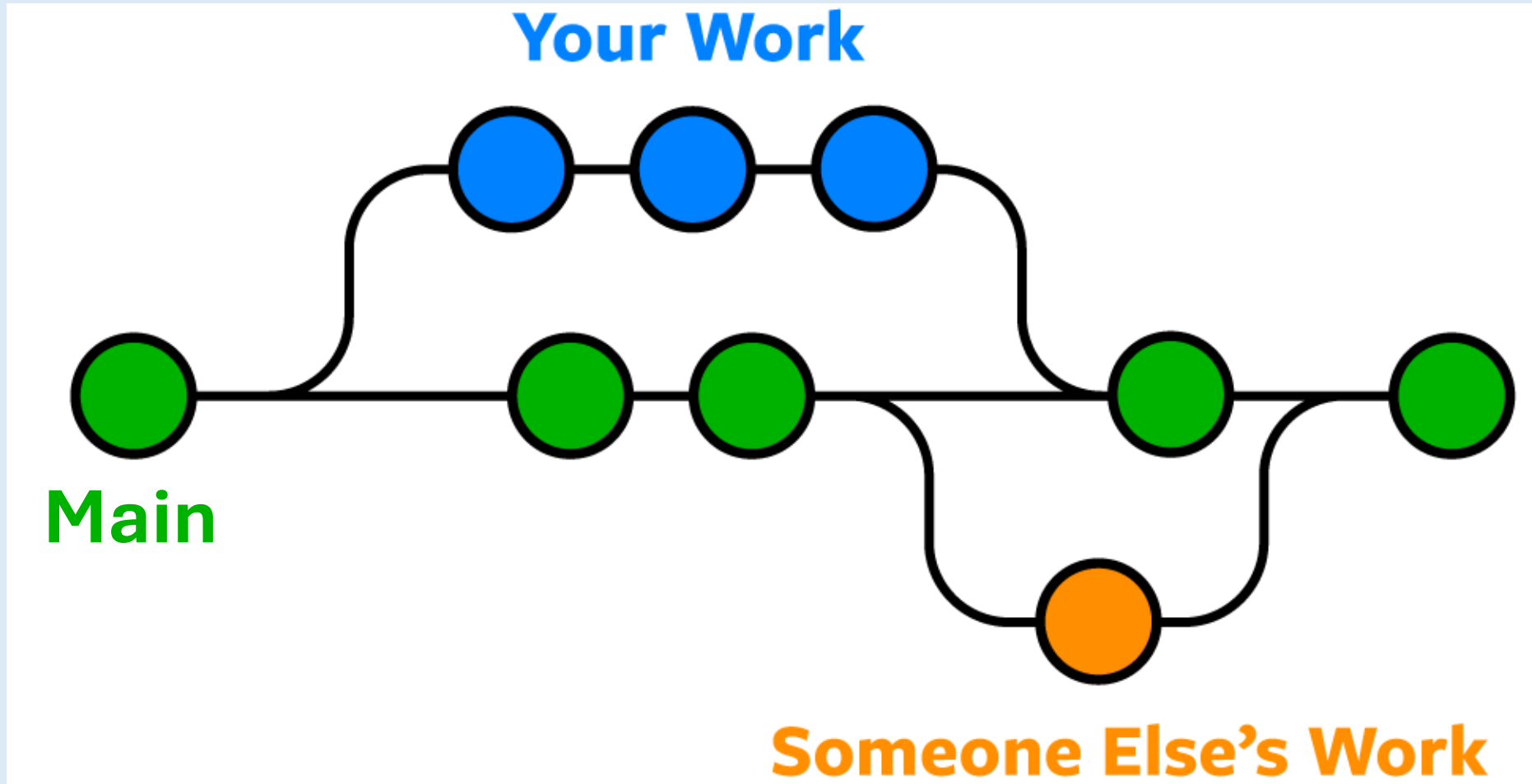
“Push code” online to save

git commit



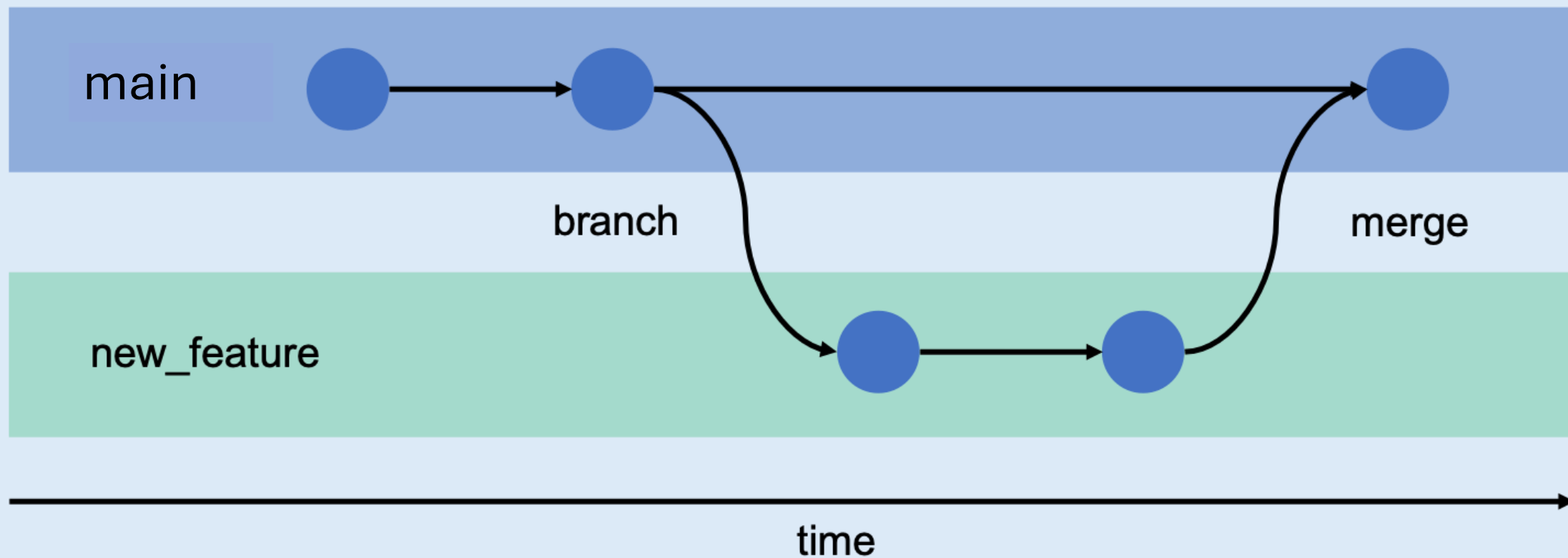
“Adding code” to “save” with a **message**

git **branches**



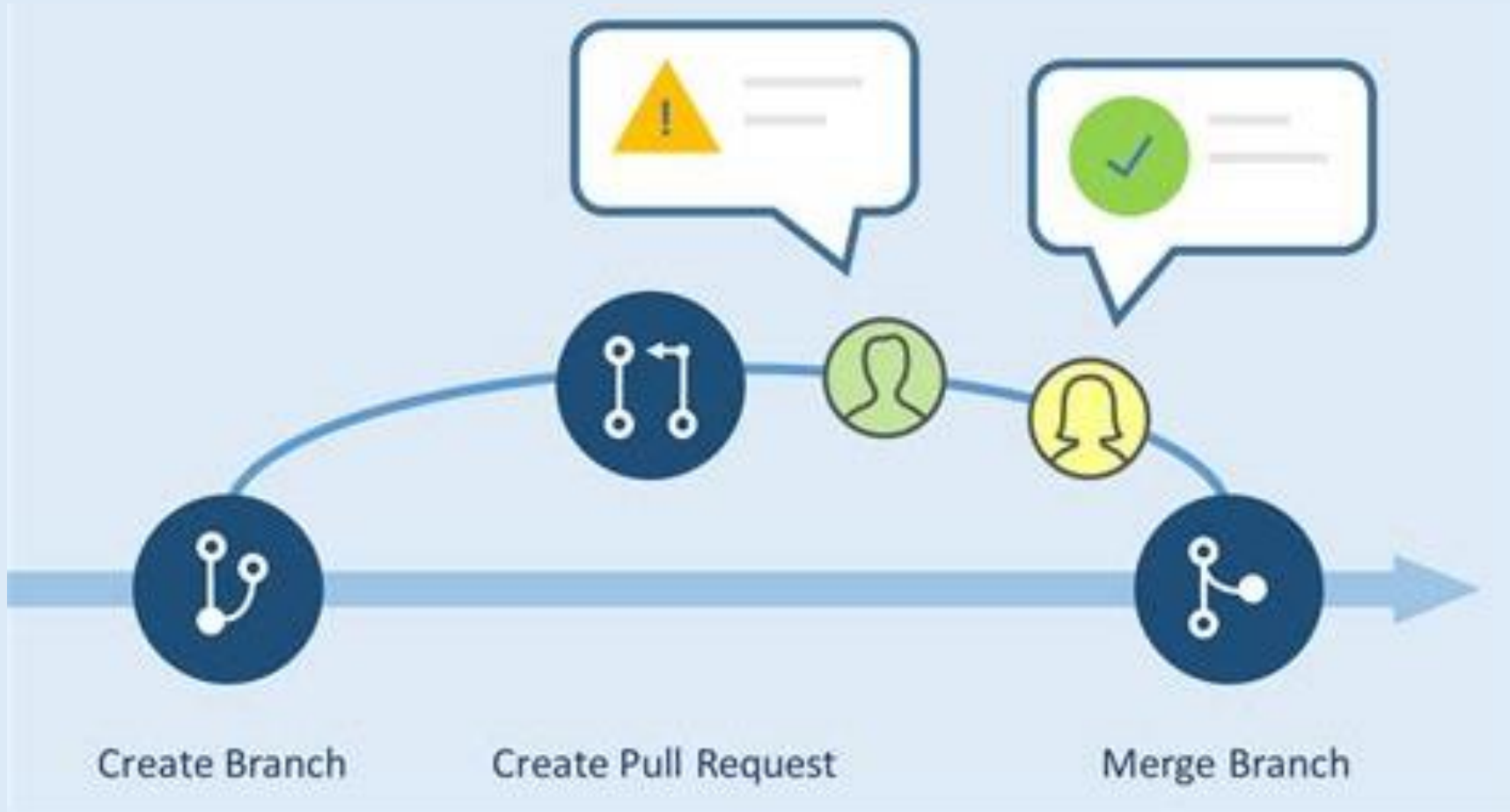
A separate **timeline**

Merging



Combining code from different branches

Pull Request



“A peer reviewed” merge

Git Collaborating

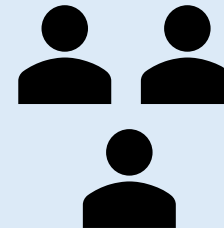
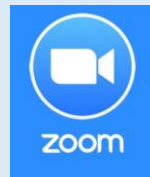
1. Collaborative Coding Sessions



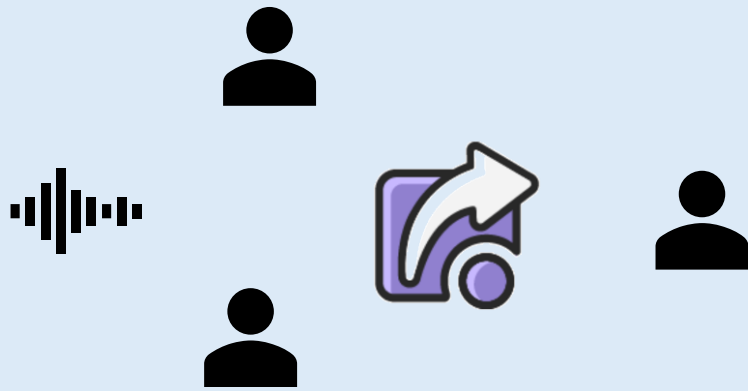
2. Individual Coding Sessions



3. Pair Coding Sessions



Collaborative Coding Sessions

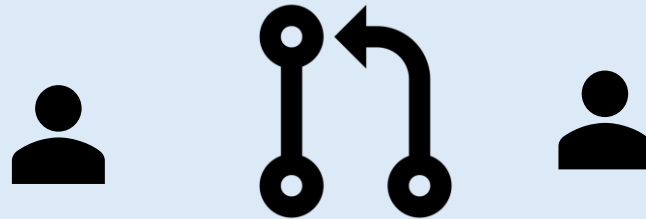


```
SESSION DETAILS
├─ Participants (3)
│   ├── Jon W Chu • Header.js:12
│   ├── Amanda Silver • GuestbookGrid.js:13
│   └── PJ Meyer • GuestbookGrid.js:9
├─ Shared Servers (2)
│   ├── localhost:3000
│   └── REST API
├─ Shared Terminals (2)
│   ├── bash (Read-only)
│   └── bash (Read/write)
└─ Audio Participants (3)
    ├── Jon W Chu
    ├── Amanda Silver
    └── PJ Meyer
```

```
1 import GridArrow from "../GridArrow";
2 import GridLegend from "../GridLegend";
3 import GuestbookGridCell from "../GuestbookGridCell";
4
5 export default class GuestbookGrid extends Component {
6   constructor(props) {
7     super(props)
8     this.state = PJ Meyer
9     signatures: signatures
10  }
11 }
12
13 Amanda Silver
14 render() {
15   const cells = this.state.signatures.map((signature, index) => (
16     <GuestbookGridCell key={index} {...signature} />
17   ));
18 }
```

Live Share extension in VS Code

Individual Coding Sessions



Pull Requests

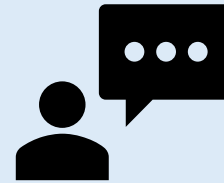
Merge changes using GitHub

Pair Coding Sessions

Look over one's shoulder

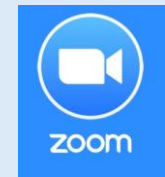


Driver



Navigator

Can happen online with screen share over Zoom/Discord/etc.



Work Side-by-side with only one person coding